The Fractional Quantum Hall Effect

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1 Introduction

The quantum hall effect (QHE) has a rich history. The Integer effect (IQHE), discovered in 1980 by Klaus von Klitzing (for which he received the 1985 Nobel Prize), is a beautiful example of a quantum system feeling topological effects. The fractional effect (FQHE) was discovered in 1982 by Horst L. Störmer and and Daniel C. Tsui, who shared the 1998 Nobel with Robert Laughlin. Laughlin's trial wave-function was able to phenomenologically describe the FQHE, but a complete microscopic description including all the observed hierarchies is still a goal of theorists today.

In the following, our primary goal is to describe the effective-field theory model proposed by Zhang, Hansson, and Kivelson [1]. Their model derives an effective theory with a Chern-Simons term (which accounts for the fractional statistics) from the microscopic Hamiltonian, and uses it to predict essentially all known aspects of the FQHE.

In the remainder of this section, we briefly review Landau levels and the phenomenology of the integer and fractional effect, and introduce Laughlin's trial wave function. In section 2, we motivate the statistical gauge field, and discuss some basic properties of Chern-Simons theory important to the FQHE. Finally, in section 3, we describe the effective theory of Zhang, Hansson, and Kivelson.

Some particularly useful references are the (older) text by Prange and Girvin [2], the more recent text by Ezawa [3], and Fradkin's book on Condensed Matter Field Theory [4]. Our discussion of Chern-Simons theory closely follows the review by Dunne [5].

1.1 The Integer Effect

We first review the classical case, following Prange [2]. We consider a 2d system, with a perpendicular magnetic field. If we run current along the plane, the current then, is given by

$$\vec{j} = \sigma_0 \vec{E} - \frac{\sigma_0}{nec} \vec{j} \times \vec{B} \tag{1}$$

where $\sigma_0 = ne^2\tau_0/m$, n the electron number density, and τ_0 the "mean free time". From this, we can read off the resistivity tensor:

$$\rho_{ij} = \begin{pmatrix} 1/\sigma_0 & B/nec \\ -B/nec & 1/\sigma_0 \end{pmatrix}_{ij} \tag{2}$$

The conductivity tensor is the matrix inverse of ρ, which now has components

$$\sigma_{xx} = \frac{\sigma_0}{1+\omega_c^2\tau_0^2}, \quad \sigma_{xy} = \frac{nec}{B} + \frac{1}{\omega_c\tau_0}\sigma_{xx} \tag{3}$$

Now we move to the quantum case. We work in the Landau gauge, where the vector potential has components $A_x = -yB$, $A_y = 0$.The Schrodinger equation then reads

$$\frac{\hbar^2}{2m}\left[\left(-i\partial_x - \frac{eB}{\hbar c}y\right)^2 - \partial_y^2\right]\psi = E\psi \tag{4}$$

If we take $\psi \sim e^{ikx}\phi(y)$, we get a shifted Harmonic oscillator potential:

$$\frac{\hbar\omega_c}{2}\left[-\ell_B^2\partial_y^2 + \left(\frac{y}{\ell_B} - \ell_B k\right)^2\right]\phi(y) = E\phi(y) \tag{5}$$

where $\ell_B^2 \equiv \hbar c/eB$ is the magnetic length. We then find eigenstates and energy levels

$$\phi_{nk}(y) \propto H_n(y/\ell_B - \ell_B k)e^{-(y-\ell_B^2 k)^2/2\ell_B^2}, \tag{6}$$

$$E_{nk} = \hbar\omega_c(n + 1/2). \tag{7}$$

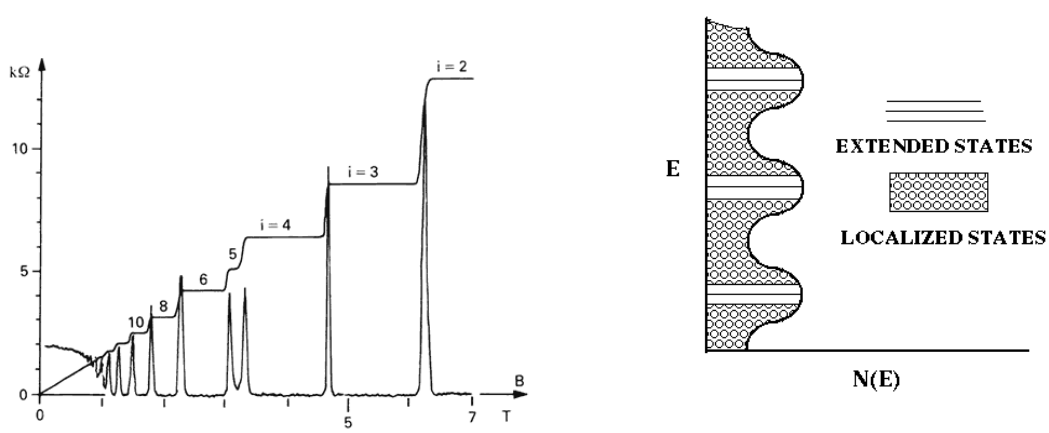
If the system has finite width, $0 < y < W$, and we impose periodic boundary conditions in x with width L, then $k = 2\pi p/L$ for some integer p, and $0 < k < W/\ell_B^2$. So, for each value of n (which defines the so-called Landau level), we have $N = LW/2\pi\ell_B^2$ degenerate states, or,

$$n_B = 1/2\pi\ell_B^2 = eB/hc \tag{8}$$

states per unit area in a Landau level. Now, if a Landau level is filled, $n = qn_B$, $q \in \mathbb{Z}$, the Fermi level lies in the gap between occupied levels, and consequently, there is no scattering: $\tau_0 \to \infty$, $1/\sigma_0 \to 0$. In this case, the inverse of the resistivity tensor becomes

$$\sigma_{ij} = \begin{pmatrix} 0 & -qe^2/h \\ qe^2/h & 0 \end{pmatrix}_{ij} \tag{9}$$

This is the integer quantum hall effect.



(a) (b)

Figure 1: Left: The original hall plateaus observed in ρ_{xy} and drops in ρ_{xx} observed by Tsui et al. [6]. Right: An illustration of the density of states of the landau levels, with impurities distinguishing extended and localized states.

However, our arguments have missed an important ingredient. Our reasoning would suggest that we should see the above conductivity quantization only at integer values of the filling fraction $\nu \equiv n/n_B$. On

the contrary, we see essentially quantized conductivity as long as ν is approximately integral (see fig. 1a). This is a result of localization - which can arise either from an applied electric field or defects. We leave the details to the references (see especially, Prange [2]), but the basic idea is that an impurity lifts the degeneracy between states in a Landau level, and hence, the ability to choose freely between extended and localized states. Instead, we find that states are extended for energies close to $q\hbar\omega_c$, and localized between these values (see Figure 1b). Only the extended states can carry current at zero temperature, giving rise to the plateaus seen in Fig. 1a.

1.2 The Fractional Effect

The fractional case has many similarities to the Integer case. Now, however, we are dealing with fractional filling – $\nu = n/n_B \in \mathbb{Q}$. Our analysis here will closely follow Fradkin [4]. For simplicity, we'll focus initially on the case $\nu = 1/m$, with m an integer. In the previous section, we saw that for $\nu < 1$ ($m > 1$) that the states in the first landau level will be degenerate, so the ground state will be some complicated linear combination of these states, likely very different from the states we found at $\nu \simeq 1$. But again, the existence of plateaus implies that an entirely different state is present. The additional piece we need is electron-electron interactions.

To describe such an effect, we need a model with $N = nA$ electrons, filling ν of the states in the lowest Landau level, interacting via the Coulomb potential. The ground state must have no gapless excitations, in order to support the plateaus, and be robust against impurities. Additionally, the wave function must be odd under interchange of any two electron coordinates so that it obeys Fermi statistics, and it must break time reversal invariance in the presence of a magnetic field. This was the program embarked upon by Laughlin [7].

The Laughlin Wavefunction In the low density limit, $\nu << 1$, the separation between electrons is much larger than the magnetic length, and the ground state is essentially a Wigner crystal, where the center coordinates defining the wave-function form a triangular lattice. However, the Wigner crystal state supports arbitrarily small excitation energy (the sound waves), and is strongly affected by impurities. As the filling fraction is increased, the ratio between the inter-electron spacing and the magnetic length approaches order unity, and we expect a phase transition to a liquid state that supports a Hall current. This liquid is dominated by quantum fluctuations, including processes involving three-particle exchanges around an elementary triangle of the lattice. The microscopic theory will reveal that there is a dynamically generated gauge field, responsible for the Higgs mechanism that will create the needed gapped excitations.

The resulting state is fundamentally different than other condensed states of matter (superconductors, magnets). To study the system, Laughlin proposed a variational wave-function:

$$\Psi_m(\vec{r}_1, \ldots, \vec{r}_N) = \frac{1}{2} \prod_{j \neq k} (z_j - z_k)^m \exp\left(- \sum_{j=1}^{N} \frac{|z_j|^2}{4\ell_B^2} \right) \tag{10}$$

The form of this wave-function is somewhat determined by demanding tha that it be odd under interchange, and be an eigenstate of the total angular momentum. For small N, Laughlin showed that the overlap between Ψ_m and the exact wave function of some generic pair interaction potentials was greater than 99%. While Laughlin originally intended Ψ_m to be a variational wave-function, the parameter m is actually determined by the total angular momentum, and thus Ψ_m doesn't contain any variational parameters. Laughlin determined this value of m by using an analogy to the probability distribution of a one-component plasma:

$$\rho(z_1, \ldots, z_N) = |\Psi_m(z_1, \ldots, z_N)|^2 \tag{11}$$

See Laughlin's article in Prange [2] for more details.

The Haldane Hamiltonians

Quasi-hole excitations of the Laughlin state There are several possible excitations of the Laughlin state - we focus on the insertion of a flux quantum, which we imagine as being created by an infinitesimally thin solenoid at one point of the sample - a Laughlin quasi-hole. For a single particle state, inserting flux ϕ at the origin changes the state from $z^n e^{-|z|^2/(4\ell_0^2)}$ to $z^{n+\phi/\phi_0} e^{-|z|^2/(4\ell_0^2)}$, with $\phi_0 = hc/e$. So, for $\phi = \phi_0$, the nth state in the first Landau level shifts to the $(n+1)$th state in the same Landau level. The same shift happens in the Laughlin state: each $z_j^{mk_j}$ is shifted to $z_j^{mk_j+1}$, with the exponential factor unchanged, and possible a change in coefficients of each z_j. This motivates the ansatz for an excited state:

$$\Psi_m^{(+)}(z_0; z_1, \ldots, z_N) = \prod_{j=1}^{N} (z_j - z_0) \Psi_m(z_1, \ldots, z_N) \tag{12}$$

Note that when any of the N electrons approaches z_0, $\Psi_m^{(+)}$ vanishes, so the charge density is depleted at z_0, which is why we think of this as a hole. A calculation using the plasma analogy (see [2]) shows that the quasi-hole has a finite excitation energy. In fact, the Laughlin state has a non-zero gap for any elementary excitation, and thus represents a uniform incompressible fluid. Another calculation (shown in Fradkin) shows that the quasi-hole also carries fractional charge e/m. Thus, electrons and holes are bound states of m fractionally charged quasiparticles.

Jain [8, 9, 10] used this construction to motivate a suggestive form of the Laughlin wave function. Write

$$\Psi_m(z_1, \ldots, z_m) = \prod_{1 \le j < k \le N} (z_j - z_k)^{m-1} \Psi_1(z_1, \ldots, z_N) \tag{13}$$

Ψ_1 is the wave function for n particles filling up the lowest Landau level, while the factor in front is interpreted as having attached a solenoid with $(m-1)$ flux quanta to each particle. This is how we can see the Laughlin state as the dynamical generation of a local gauge field (the solenoids) which partially screens the external magnetic flux. The screening is precisely enough to turn the fractional filling of the Landau level of the bare field into the complete filling of a Landau level of the un-screened part of the field. This is the starting point of the field-theoretic description of the field.

2 Chern-Simons Theory

Chern-Simons theory is an important part of many branches of theoretical physics, and has many beautiful mathematical properties. Here, we'll focus on the Abelian Chern-Simons theory, in 2+1 dimensions, following [5], but non-Abelian generalizations to any odd number of dimensions exist as well. The Chern-Simons Lagrangian we consider here takes the form

$$\mathcal{L}_{CS} = \frac{\theta}{2} \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho - A_\mu J^\mu. \tag{14}$$

Note that although $\mathcal{L}_{CS}$ depends on A_μ explicitly, it's variation under a gauge transformation is a total spacetime derivative:

$$\delta \mathcal{L}_{CS} = \frac{\theta}{2} \partial_\mu \left(\lambda \epsilon^{\mu\nu\rho} \partial_\nu A_\rho \right). \tag{15}$$

So the action is gauge invariant, provided we can neglect boundary terms[1].

[1] Of course there are many interesting cases where the boundary terms cannot be neglected, and they give rise to many interesting phenomena in their own right, but we won't discuss those here.

The classical field equation for A_μ is

$$F_{\mu\nu} = \frac{1}{\theta}\epsilon_{\mu\nu\rho}J^\rho \tag{16}$$

In terms of components, (recall that in 2+1 dimensions, $F_{\mu\nu}$ has only 3 degrees of freedom, and the "magnetic field" has only one component):

$$\rho = \theta B \tag{17}$$

$$J^i = \theta\epsilon^{ij}E_j \tag{18}$$

These have a clear interpretation, relevant to the FQHE: the charge density is locally proportional to the magnetic field. In other words, a Chern-Simons term attaches magnetic flux to charged particles - precisely the interpretation we gave to Jain's formulation of the Laughlin wave-function. If we consider particles with charge e, the charge density is given by

$$\rho(\vec{x},t) = e\sum_{a=1}^{N}\delta(\vec{x}-\vec{x}_a(t)) \tag{19}$$

If we add a kinetic term for these particles to the action and solve for the equations of motion in a Hamiltonian formulation, we can gauge-fix by setting $A_0 = 0$ and $\vec{\nabla}\cdot\vec{A} = 0$. Then,

$$A^i(\vec{x},t) = \frac{1}{2\pi\theta}\int d^2y\epsilon^{ij}\frac{(x^j-y^i)}{|\vec{x}-\vec{y}|^2}\rho(\vec{y},t) = \frac{e}{2\pi\theta}\sum_{a=1}^{N}\epsilon^{ij}\frac{(x^j-x_a^j(t))}{|\vec{x}-\vec{x}_a(t)|^2} \tag{20}$$

Now, the Hamiltonian for the point particles is

$$H = \frac{1}{2m}\sum_{a=1}^{N}\left(\vec{p}_a - e\vec{A}(\vec{x}_a)\right)^2 \tag{21}$$

Where $\vec{A}(\vec{x}_a)$ is as above, with the sum excluding the particle's own field. So, each particle sees a magnetic field

$$B(\vec{x}_a) = \frac{e}{\theta}\sum_{b\neq a}^{N}\delta(\vec{x}_a-\vec{x}_b). \tag{22}$$

In other words, the other particles look like point vortices with flux $\Phi = e/\theta$. This first-quantized picture is enough to see the anyonic statistics of these charged particles. If we adiabatically "move" one particle around another, the non-relativistic wave-function acquires an Aharonov-Bohm phase

$$\exp\left(ie\oint_C \vec{A}\cdot d\vec{x}\right) = \exp\left(\frac{ie^2}{\theta}\right) \tag{23}$$

Which corresponds to an "exchange" phase $2\pi\delta = e^2/2\theta$. This is a realization of the notion that, in two-dimensions, we are no longer restricted to particles having exchange phases of ± 1, but they can in fact take any value [11]. This is rooted in spin-statistics, and that only integer and half-integer spin representations exist for the group of rotations $SO(3)$, while $SO(2)$ is Abelian, and allows arbitrary spin representations.

So, we should interpret the Chern-Simons theory as a constraining theory that ties magnetic flux to particles. This can be put in a field theoretic framework as well. This framework allows us to write a theory of anyons with statistical phase δ in terms of a theory of fermions coupled to a Chern-Simons gauge field with coupling $\theta = 1/(2\delta)$. Alternatively, we can represent the anyonic theory as a theory of *bosons* coupled to a Chern-Simons gauge field but with coupling $\theta = 1/(2(\delta\pm\pi))$, provided that the bosons have "hard cores" (see [4]) for details.

3 Effective Field Theory

With the tools of Chern-Simons theory at our disposal, we can now derive Zhang, Hansson, and Kivelson's effective field theory for the fractional quantum Hall effect[1]. Their approach was to derive essentially a Landau-Ginzburg theory - valid at low energies and long distances - capable of predicting most of the qualitative features of the FQHE.

We consider a system of N electrons moving on a plane in the presence of an external uniform magnetic field perpendicular to the plane. We assume that the magnetic field is strong enough that all the electrons are polarized and we can ignore their spin degree of freedom. The electrons are subject to a pairwise interaction V, and hence, we have the Hamiltonian

$$\hat{H} = \sum_{i=1}^{N} \left\{ \frac{1}{2m} \left(\vec{p}_j - \frac{e}{c} \vec{A}_j(\vec{r}_j) \right)^2 + eA^0(\vec{r}_j) \right\} + \frac{1}{2} \sum_{i \neq j} V(|\vec{r}_i - \vec{r}_j|) \tag{24}$$

In second quantized notation, this becomes

$$\mathcal{S} = \int d^2r \left[\psi^*(r) i D_0 \psi(r) + \frac{\hbar^2}{2m} |\vec{D}\psi(r)|^2 \right] \tag{25}$$

$$-\frac{1}{2} \int d^2r \int d^2r' \left(|\psi(r)|^2 - \rho_0 \right) V(|r - r'|) \left(|\psi(r')|^2 - \rho_0 \right) \tag{26}$$

where ρ_0 is the average density, and D_μ is the covariant derivative in the presence of a background gauge field, A. Now, we map this to a system of (composite) bosons coupled to a "statistical gauge potential" $a(r)$. This gives us (in the notation of ref. [1])

$$H = \int d^2r \phi^*(r) \left[\frac{1}{2m} \left(-i\vec{\nabla} - e\vec{A}(r) - e\vec{a}(r) \right)^2 + eA^0(r) \right] \phi(r) \tag{27}$$

$$+\frac{1}{2} \int d^2r d^2r' \phi^*(r) \phi^*(r') V(r - r') \phi(r) \phi(r') \tag{28}$$

where the statistical gauge field is given by (see eq. (20))

$$a^i(r) = \frac{\theta}{\pi e} \epsilon^{ij} \int d^2r' \frac{r_j - r'_j}{|r - r'|^2} \phi^*(r') \phi(r') \tag{29}$$

where the Chern-Simons coupling in this mapping is chosen to be an arbitrary odd-integer multiple of π: $\theta = (2k - 1)\pi$. Because of the coupling to $a(r)$, $\phi(r)$ is an *anyonic* field - under interchange of particles, the wave-function of ϕ changes by a phase θ. The "statistical gauge field"[2] b is given by

$$b(r) = -\epsilon^{ij} \partial_i a_j(r) = \frac{2\theta}{e} |\phi(r)|^2 \equiv s|\phi(r)|^2 \left(\frac{2\pi}{e} \right) \tag{30}$$

This is a constraint equation - we incorporate it into the action through a Lagrange multiplier field, which we can conveniently label a_0. Now, we have a coherent-state path-integral representation for the quantum partition function:

$$Z[A^\mu] = \int [d\phi][da_i^T][da_0] e^{iS[\phi, a_i^T, a_0]} \tag{31}$$

where the Lagrangian is

$$\mathcal{L} = i\phi^* \partial_0 \phi - H(\phi) + \mu \phi^* \phi - a_0 [\frac{e^2}{2\theta} \epsilon^{ij} \partial_i a_j^T + e\phi^* \phi] \tag{32}$$

[2] I'm using the terminology of [1] here – perhaps a better convention is that a_μ is the "statistical gauge field", while b given here is the "statistical magnetic field"

The last piece is the Chern-Simons term in a particular gauge - we can return to the usual form by "reversing the Fadeev-Popov procedure". Before we can extract the long distance behavior, we have to make a couple approximations. First, in order for the mapping onto bosonic fields to make sense, the bosons must have a "hard core". Since we're only interested in the long-distance behavior, we can replace this by a short-distance repulsive force, which amounts to adding a term $-\lambda|\phi(r)|^4$ to the Lagrangian. For simplicity, we also absorb the electron repulsive force into λ. We also replace the bare mass $1/m$ with a renormalized "stiffness" constant κ. Finally, we arrive at the effective Lagrangian

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_\phi + \mathcal{L}_{CS} && (33)\\ &= \phi^*[i\partial_0 - e(A_0+a_0)]\phi - \frac{1}{2}\kappa\phi^*[-i\vec{\nabla} - e(\vec{A}+\vec{a})]^2\phi + \mu|\phi|^2 - \lambda|\phi|^4 - \frac{e^2}{4\theta}\epsilon^{\mu\nu\sigma}a_\mu\partial_\nu a_\sigma && (34)\end{aligned}$$

This completes the Zhang, Hansson, and Kivelson's derivation of the effective action for the FQHE. Now, we show that this model reproduces the important properties of the FQHE.

3.1 Hall Conductance

Before computing the Hall conductance, it is useful to look at the classical equations of motion we get from (33). Varying with respect to ϕ^*, a_0, a_i respectively, we find

$$\begin{aligned}(i\partial_0 - e(A_0+a_0))\,\phi - \frac{\kappa}{2}\left(-i\vec{\nabla} - e(\vec{A}+\vec{a})\right)^2\phi + \mu\phi - \lambda\phi|\phi|^2 &= 0 && (35)\\ |\phi|^2 + \frac{e}{\theta}b(r) &= 0 && (36)\\ \frac{e^2}{\theta}\epsilon^{i\alpha\beta}\partial_\alpha a_\beta + \frac{\kappa}{2}\left[\phi^*\left(i\vec{\nabla} + e(\vec{A}+\vec{a})\right)\phi - \left(\left(i\vec{\nabla} + e(\vec{A}+\vec{a})\right)\phi^*\right)\phi\right] &= 0 && (37)\end{aligned}$$

We also work with constant density, so $|\phi|^2 = n$. It's clear that the ground state will be stationary, and have constant amplitude which reduces the above to

$$\begin{aligned}\mu - 2\lambda n &= 0 && (38)\\ n + \frac{e}{\theta}\langle b\rangle &= 0 && (39)\\ a_\mu + A_\mu &= 0 && (40)\\ \frac{e^2}{\theta}\epsilon^{i\alpha\beta}\partial_\alpha a_\beta &= 0 && (41)\\ & && (42)\end{aligned}$$

The third of these shows that the average statistical gauge field screens the background electromagnetic field, and hence, for a constant background magnetic field B, we get $\langle b\rangle = -B$. But then, the second equation above tells us that the particle density will be proportional to B: $n = eB/\theta$. But then, the filling fraction $\nu = n/(eB) = \pi/\theta$. So clearly, this solution exists only for the particular fraction determined by θ[3]

Now we compute the Hall conductance. To do so, we apply an external scalar potential (electric field) $E_i = -\partial_i A_0$, in addition to a constant magnetic field $B = -\epsilon^{ij}\partial_i A_j$. Now, we compute the current in a state satisfying the equations of motion. We have

$$j^i = \frac{\delta S}{\delta A_i} = -\frac{\delta S_\phi}{\delta a_i} \qquad (43)$$

[3]As pointed out in [1], there could also be solutions with other fractions, related by singular gauge transformations.

Where S_ϕ is the part of the Lagrangian depending on ϕ, and we've used the equation of motion to set $A_i = -a_i$. But since we're looking at solutions of the equations of motion, $\delta S/\delta a_i = 0$, so

$$-\frac{\delta S_\phi}{\delta a_i} = +\frac{\delta S_{CS}}{\delta a_i} = \frac{e^2}{2\theta}\epsilon^{ij}(\partial_0 a_j - \partial_j a_0). \tag{44}$$

where S_{CS} is the Chern-Simons term in the effective action. Now, we just have to compute

$$\langle j^i \rangle = \frac{1}{Z}\int [d\phi da_\mu]\ \frac{e^2}{2\theta}\epsilon^{ij}(\partial_0 a_j - \partial_j a_0) e^{iS[\phi, a_0, a_i, A_0, A_i]} \tag{45}$$

So, using the equations of motion again,

$$\langle j^i \rangle = -i\frac{\delta \log Z}{\delta A_i}\bigg|_{A_0=0, A_i}^{A_0, A_i} \tag{46}$$

Now, we integrate out the statistical gauge field, and expand the action to lowest order in $A_\mu + \delta A_\mu$. The first order contribution comes from the Chern-Simons term:

$$S_{eff}[A_\mu + \delta A_\mu] = \frac{e^2}{4\theta}\int d^3r \epsilon^{\mu\nu\sigma}\delta A_\mu \partial_\nu A_\sigma + \dots \tag{47}$$

Now, the spatial component of the current comes from the variational derivative of this with respect to A_i, so we find

$$\langle j^i \rangle = -\frac{e^2}{2\theta}\epsilon^{ij}\partial_j A_0 = \frac{e^2}{2\theta}\epsilon^{ij}E_j \tag{48}$$

In other words, we have a Hall conductivity:

$$\sigma_H^{ij} = \frac{e^2}{2\theta}\epsilon^{ij} = \frac{e^2}{2\pi(2k-1)}\epsilon^{ij}. \tag{49}$$

So we indeed recover the fractionally quantized Hall conductivity.

3.2 Vortex Excitations

The equations of motion above (35-37) admit static, non-uniform vortex solutions with finite energy [1] Using polar coordinates, with the center of the vortex at $r = 0$, we have at large r,

$$\phi(r,\varphi) = \sqrt{n}e^{\pm i\varphi} \tag{50}$$

$$\vec{a}(r,\varphi) = \pm\hat{\varphi}/er \tag{51}$$

$$a_0(r,\varphi) = 0 \tag{52}$$

These solutions correspond to one unit of statistical flux per vortex. We can use the equations of motion to compute the charge carried by each vortex, using a similar trick as before:

$$j^0 = -\frac{\delta S}{\delta A_0} = -\frac{\delta S_\phi}{\delta a_0} = +\frac{\delta S_{CS}}{\delta a_0} = \frac{e^2}{2\theta}b(r) \tag{53}$$

With $b(r)$ the "statistical magnetic field" from before. Now using Gauss' law:

$$q \equiv \int d^2r j^0 = \frac{-e^2}{2\theta}\int d^2r\epsilon^{ij}\partial_i a_j(r) = \frac{e^2}{2\theta}\oint \vec{a}\cdot d\vec{r} = \pm\frac{\pi}{\theta}e = \pm\frac{e}{2k-1} \tag{54}$$

So we see that the vortex excitations are fractionally charged, just as the quasi-particle excitations in the Laughlin scheme were. It can be shown [12, 11, 13, 14, 15] that this also implies the quasiparticles obey fractional statistics, just as Laughlin's quasiparticles. These excitations are the starting point to the hierarchy scheme of Haldane and Halperin ([16, 17]), used to explain the other observed filling fractions.

3.3 The Higgs Mechanism and Collective Excitations

Now we consider collective excitations. Note that the sign of the chemical potential μ in $\mathcal{L}_\phi$ allows for spontaneous symmetry breaking (we saw this explicitly earlier when we computed the constant amplitude solutions to the equations of motion). So, the field ϕ acquires a vacuum expectation value $|\langle\phi\rangle| = \sqrt{n}$, and we can parametrize

$$\phi(r) = (\phi_0 + \delta\phi(r))\, e^{ie\eta(r)} \tag{55}$$

But then, the statistical gauge field can be parametrized

$$a_\mu(r) = \delta a_\mu(r) + \partial_\mu \eta(r) \tag{56}$$

And we see that we can gauge away all the phase fluctuations, $\eta(r)$, via the Anderson-Higgs mechanism, so there are no gapless excitations, as mentioned in the first section.

4 Conclusions

We've successfully developed an effective "Landau-Ginzburg" description of the fractional quantum hall effect. Of course, there's a lot we've left out. One of the next developments historically, was the fermionic theory developed by Lopez and Fradkin [18].

We've also barely scratched the surface of Chern-Simons theory. A fuller treatment would discuss the braiding statistics more in detail (see [5, 4], for starters). Further generalizations of Chern-Simons theory have wound their way into a variety of places in modern physics - including String theory. Additionally, we've left out a discussion of boundaries throughout - the boundary of a Chern-Simons theory is related to a Wess-Zumino-Witten (WZW) model - a 2-dimensional conformal field theory, and there is a great deal of real phenomena at the boundary of a quantum hall system that can be described in this way.

Finally, I wanted to mention one of the more recent works that motivated me to learn more about the conventional FQHE. Vafa [19] has recently noted an interesting connection between the FQHE and recent developments in string theory, that also makes different predictions for the observed statistics of the quasiholes in the principal series of the FQHE. Of course, the necessary background for discussing such a paper here would take us far afield.

References

[1] S C Zhang, T H Hansson, and S Kivelson. Effective-Field-Theory Model for the Fractional Quantum Hall Effect. *Physical Review Letters*, 62(1):82–85, jan 1989.

[2] Richard E. Prange and Steven M. Girvin. *The Quantum Hall Effect*. Springer-Verlag, New York, 1987.

[3] Z F Ezawa. *Quantum Hall effects: field theoretical approach and related topics*. World Scientific, 2nd edition, 2008.

[4] Eduardo Fradkin. *Field Theories of Condensed Matter Physics*. Cambridge University Press, 2nd edition, 2013.

[5] Gerald V. Dunne. Aspects of Chern-Simons Theory. 1999.

[6] D. C. Tsui, H. L. Stormer, and A. C. Gossard. Two-dimensional magnetotransport in the extreme quantum limit. *Physical Review Letters*, 48(22):1559–1562, 1982.

[7] R. B. Laughlin. Anomalous Quantum Hall Effect: An Incompressible Quantum Fluid with Fractionally Charged Excitations. *Physical Review Letters*, 50(18):1395–1398, may 1983.

[8] J. K. Jain. Composite-fermion approach for the fractional quantum Hall effect. *Physical Review Letters*, 63(2):199–202, 1989.

[9] J. Jain. Incompressible quantum Hall states. *Physical Review B*, 40(11):8079–8082, 1989.

[10] J. Jain. Theory of the fractional quantum Hall effect. *Physical Review B*, 41(11):7653–7665, 1990.

[11] Frank Wilczek. Quantum Mechanics of Fractional-Spin Particles. *Physical Review Letters*, 49(14):957–959, oct 1982.

[12] Daniel Arovas, J. R. Schrieffer, and Frank Wilczek. Fractional Statistics and the Quantum Hall Effect. *Physical Review Letters*, 53(7):722–723, aug 1984.

[13] Daniel P. Arovas, Robert Schrieffer, Frank Wilczek, and A. Zee. Statistical mechanics of anyons. *Nuclear Physics B*, 251:117–126, jan 1985.

[14] T.H. Hansson, M. Roček, I. Zahed, and S.C. Zhang. Spin and statistics in massive (2+1)-dimensional QED. *Physics Letters B*, 214(3):475–479, nov 1988.

[15] Gordon W. Semenoff. Canonical quantum field theory with exotic statistics. *Physical Review Letters*, 61(5):517–520, 1988.

[16] F. D. M. Im Haldane. Fractional Quantization of the Hall Effect: A Hierarchy of Incompressible Quantum Fluid States. *Physical Review Letters*, 51(7):605–608, aug 1983.

[17] B. I. Halperin. Statistics of Quasiparticles and the Hierarchy of Fractional Quantized Hall States. *Physical Review Letters*, 52(18):1583–1586, apr 1984.

[18] Ana Lopez and Eduardo Fradkin. Fractional quantum Hall effect and Chern-Simons gauge theories. *Physical Review B*, 44(10):5246–5262, sep 1991.

[19] Cumrun Vafa. Fractional Quantum Hall Effect and M-Theory. nov 2015.